

IDS Platform — System Design

Intrusion Detection System • Flask API • Angular • MySQL • ML/DL

Network Intrusion Detection System

A full-stack real-time IDS platform powered by a trained ML/DL model (XGBoost / CNN), a Flask REST API backend, a MySQL database for logging, and an Angular dashboard for live monitoring, alerting and explainability.

End-to-End Architecture

System Architecture — End-to-End IDS Platform



Best Model XGBoost /
CNN

Primary Metric Balanced
Accuracy

Features Used 34 (after
corr. drop)

Dataset CIC-IDS-2018
16M rows

1 Angular Dashboard — Pages & Features

The Angular frontend consists of five main pages, each serving a specific role in the IDS workflow — from live monitoring to batch analysis and model transparency.

● Live Monitor

- Real-time flow feed (auto-refresh)
- Green = Benign | Red = Attack
- Confidence % per prediction

▲ Alerts

- Attack-only filtered view
- Severity: High / Medium / Low
- Mark as reviewed / resolved

■ Statistics

- Attack rate today vs yesterday
- Hourly traffic line chart
- Benign vs Attack pie chart

◆ Upload & Analyze

- Upload CSV of network flows
- Batch prediction on all rows
- Download results as CSV/PDF

★ Model Info

- Active model name + version
- Balanced Accuracy / F1 Macro
- Benign & Attack Recall shown

✦ Settings / Auth

- JWT login (Admin / Analyst)
- Role-based access control
- Alert threshold configuration

2 Key Platform Features

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SHAP ■ Explainability

Top 5 features that triggered each attack prediction with bar chart

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Confidence ■ Scoring

Model probability 0-100%. Low confidence flagged as Needs Review

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IP Reputation ■ Tracking

Flags IPs that repeatedly trigger alerts within 24h window

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Batch CSV ■ Upload

Analyze entire capture files. Downloadable PDF/CSV report output

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JWT Auth ■ & Roles

Admin, Analyst, Viewer roles. Secure token-based authentication

3 MySQL Database Schema

Five tables cover the full data lifecycle — from raw predictions to aggregated statistics, alerts, user management and model registry.

predictions	alerts	traffic_stats	users	models
id (PK)	id (PK)	id (PK)	id (PK)	id (PK)
timestamp	timestamp	hour	username	name
src_ip	src_ip	total_flows	password_hash	version
dst_ip	dst_ip	attack_count	role	bal_accuracy
prediction	confidence	benign_count	last_login	f1_macro
confidence	severity	avg_confidence		file_path
model_used	status			is_active
features_json				

Note: Yellow-highlighted rows indicate primary keys. The *predictions* table stores every flow analysed. The *models* table enables future model versioning and switching.

4 Flask REST API Endpoints

All endpoints are prefixed with **/api**. Prediction endpoints require a valid JWT token. Auth endpoints are public.

Method	Endpoint	Description
POST	/api/auth/login	Authenticate user, return JWT token
POST	/api/predict	Single flow prediction — returns label + confidence
POST	/api/predict/batch	CSV upload — batch prediction on all rows
GET	/api/alerts	Get all alerts (filter by status/severity/date)
PUT	/api/alerts/:id	Mark alert as reviewed or resolved
GET	/api/stats/live	Live dashboard counters (total, attacks, rate)
GET	/api/stats/hourly	Hourly traffic data for line chart
GET	/api/model/info	Active model name, version, metrics, features
GET	/api/predict/explain/:id	SHAP explanation for a specific prediction
GET	/api/export/report	Download predictions as CSV or PDF report

5 Technology Stack

Each layer is independently deployable. The ML layer loads the trained model once at startup and serves predictions in memory — no retraining at runtime.

Frontend Layer

Angular 17 • Chart.js • ngx-charts • Socket.IO

API Layer

Flask • Flask-JWT • Flask-CORS • SQLAlchemy

ML/DL Layer

XGBoost • TensorFlow/Keras • Scikit-learn • SHAP

Data Layer

MySQL • Pandas • NumPy

Infrastructure

Python 3.11 • Docker (optional) • REST / WebSocket

6 SHAP Explainability — Why This Prediction?

SHAP (SHapley Additive exPlanations) explains *why* the model classified a flow as an attack. For each prediction flagged as Attack, the platform shows the top 5 features that contributed most to the decision.

Feature	Direction	Impact	Explanation
Flow Bytes/s	HIGH ↑	STRONG	Abnormally high byte rate — typical of DoS flood
Bwd IAT Min	LOW ↓	STRONG	Very short inter-arrival time — rapid repeated packets
Init Bwd Win Bytes	LOW ↓	MEDIUM	Small window size — scanning or malformed handshake
Fwd Pkt Len Mean	HIGH ↑	MEDIUM	Large average packet size — data exfiltration pattern
Flow Duration	LOW ↓	WEAK	Very short flow — port scan or connection probe

7 What Makes This Project Stand Out

Feature	Why It Impresses
SHAP per-prediction explanation	Shows deep ML understanding beyond just accuracy numbers
Balanced Accuracy as primary metric	Honest evaluation on imbalanced dataset (not inflated accuracy)
8 ML + 2 DL models compared	Systematic evaluation — not just picking one model and hoping
Confidence score + Needs Review tier	Handles model uncertainty in a production-aware way
JWT role-based access	Shows backend security awareness (Admin / Analyst / Viewer)
Batch CSV upload + PDF export	Real-world usability beyond just a live demo
MySQL audit trail	Every prediction logged — traceable and auditable
Hourly attack heatmap	Data engineering thinking, not just ML notebook work